

Varun Pulipati

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Data Scientist with 4+ years building predictive models and ML systems in production across fraud detection, loyalty analytics, and published research. Strong in Python, SQL and translating model outputs into business decisions. Consistent builder of self-started tools and creative solutions from open-source libraries to LLM-powered agents

EXPERIENCE

Data Scientist | RaceTrac - Atlanta, GA

Jul 2024 - Present

- Quantified a **retention gap shaping the re-engagement roadmap** by owning measurement strategy for loyalty platform migration serving 5M+ members, defining retention KPIs, and building SQL/Python cohort reporting
- Prioritized **top 15% at-risk members for targeted campaigns** by building an end-to-end churn prediction system (gradient-boosted trees): feature engineering, lift-by-decile evaluation, and A/B holdout design
- Enabled **leadership to track Loyalty ROI against forecast** by developing a Loyalty Goals Tracker using Prophet/ARIMA with rolling backtesting and confidence intervals to compare forecast vs. actual performance
- Reduced **ad-hoc requests by 50%** by consolidating 60+ raw loyalty data tables into 20 curated Delta tables (Databricks/PySpark) and deploying AI/BI Genie self-serve layer
- Contributed to a **multi-agent AI initiative** that converts meeting transcripts and notes into end-to-end delivery outputs (requirements, solution design, dev tasks, QA test cases) surfaced through a business facing **Azure AI Foundry** page.

Research Scientist | University of Missouri (NSF) - Columbia, MO

Aug 2022 - May 2024

- Quantified **physical demand in real time** by designing a platform that unified data from multiple AR/motion-tracking SDKs
- Achieved **80% precision in identifying high-fatigue states** by implementing an ML fatigue scoring model (0-100 index) using regression on motion-capture features
- Automated the **full ML pipeline end-to-end** with self-hosted Airflow (OneDrive-triggered DAGs) to clean motion data, engineer features, predict fatigue, and publish results to Tableau dashboards
- 3 first-author publications and Best Student Paper Award 2024** in Ergonomics journal, HCII, and HFES conferences

Software Engineer Fraud Analytics | Accenture (Aetna/CVS Health)

Aug 2020 - Dec 2021

- Improved early **fraud intervention 20%** by developing XGBoost fraud detection models on 2M+ healthcare claims, engineering features from provider billing anomalies and temporal claim patterns, and tuning thresholds with fraud-ops
- Cut pipeline **delivery from 5 days to 24 hrs** by building automated SQL-driven data pipelines for daily claims ingestion
- Enabled **fraud-ops to prioritize high-confidence cases** by creating Power BI dashboards tracking false-positive rates and investigation outcomes to measure investigation ROI

PROJECTS

[Speak2py](#) (Python Library)

Jun 2025

- Open-source library converting plain-English data queries into executable Pandas/SQL via an **offline LLM in a sandboxed environment**. Features schema-aware prompt engineering, automatic code validation, and an eval harness benchmarking offline vs. cloud LLM accuracy

[Unloop AI](#) (Recommendation Algorithm)

Nov 2025

- Unloop AI is a trajectory based **recommendation engine** that matches a user's current behavior signals to other users past states and recommends what they evolved into next. It turns browsing events into weekly feature vectors and ranks candidate evolution paths using similarity and velocity to produce explainable, time aware recommendations

PriceLift Lab (Elasticity + Experimentation Framework)

Jan 2026

- Projected **\$1.2M annual revenue lift** by building a Python pricing pipeline estimating elasticity across 8 categories (log-log + DiD) and a revenue model
- Validated results on a **100K-user experiment** using an end-to-end A/B testing with power analysis, SRM checks, and guardrails

TECHNICAL SKILLS

- Data Science & Analytics:** Problem Framing, EDA, Statistical Modeling, Regression/Classification, Feature Engineering, Clustering, Time-Series Forecasting (Prophet, ARIMA), Causal Inference (PSM, DiD), Experiment Design & A/B Testing
- Programming & Machine Learning:** Python (Pandas, NumPy, SciPy, PySpark), SQL, Scikit-learn, TensorFlow, XGBoost, PyTorch, Neural Networks, NLP/LLMs, REST APIs (FastAPI, Flask), Testing (Pytest, Postman), Git
- Data Platforms & Engineering:** Azure Databricks, ETL/ELT, Spark, Airflow, Relational & NoSQL databases, Snowflake, BigQuery, Data Warehousing, Data Governance & Data Quality
- Cloud & MLOps:** Azure (ADLS, ADF, ADO), AWS (S3/Redshift), Model Monitoring (PSI, KS), CI/CD, MLflow, Delta Lake
- Visualization & BI:** Power BI (DAX), Tableau, Looker, Streamlit, Logging & Observability

EDUCATION

M.S. Computer Science, University of Missouri, Columbia

USA (May 2024)

B.Tech Electronic and Computer Engineering, CVR College of Engineering

India (May 2020)

ACHIEVEMENTS & PUBLICATIONS

- Impact Award (2025) recognized by Enterprise Director and business partners
- Publications (8) and Best Student Paper Award (HFES, 2024)